

Break Even Point (BEP)

- Basic tools for CVP analysis
- **CVP analysis** – the total cost of a firm depends on its volume of output. The total cost has two parts : fixed and variable. Fixed remains fixed at all levels of production in the short-run but variable cost proportionately changes with the volume of output. Again profit of the firm is dependent on its total sales and total cost. It is because the positive difference between gross revenue and total cost is profit. So there is a direct relationship among cost of production, volume of output and profit earned.
- **BEP** is the level of sales at which revenue equals costs and net income is zero.
- It is also known as no profit no loss point because at this point sales revenue and cost are equal.

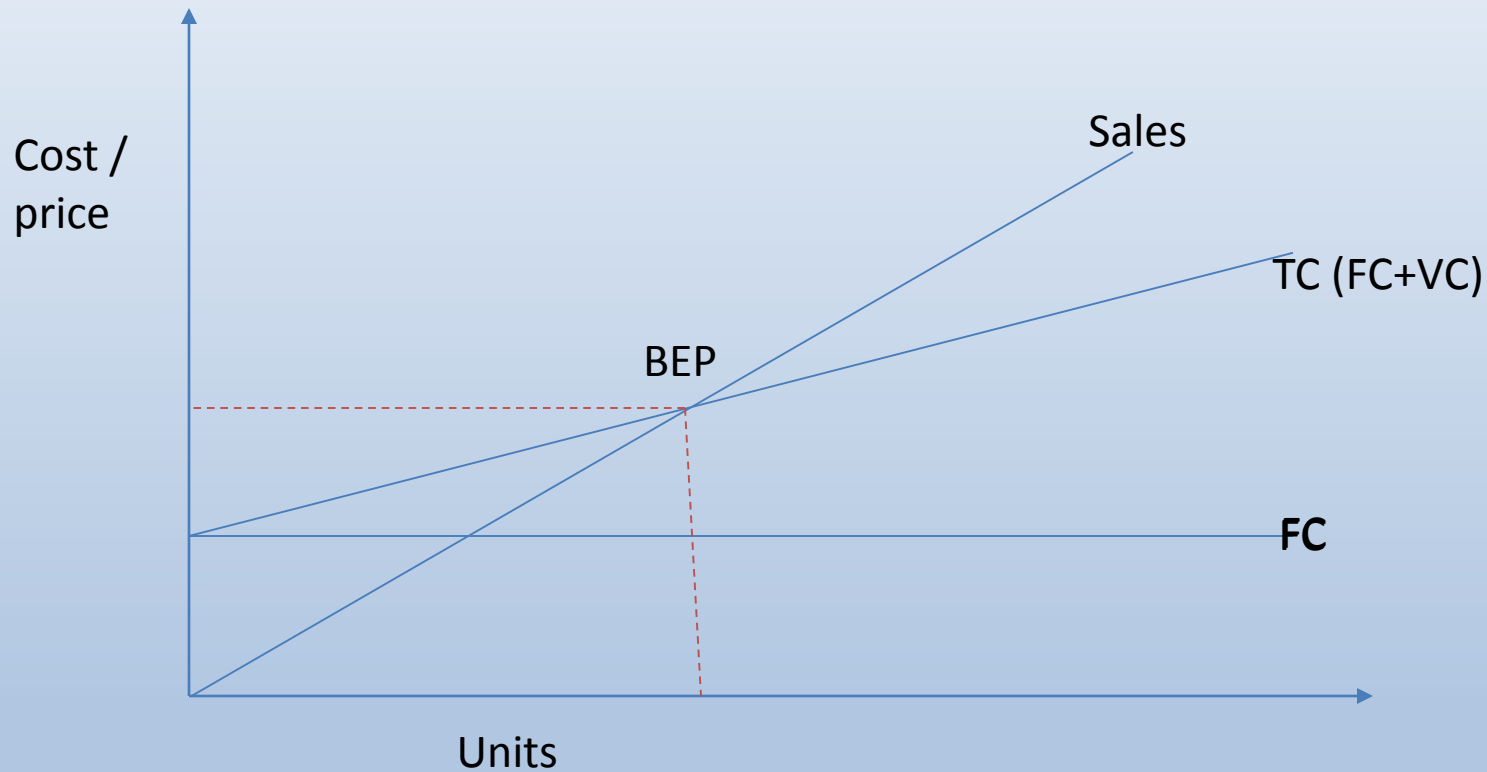
Importance of BEP

- Profit planning.
- Production planning.
- Taking pricing decision.
- Achieving efficiency.
- Help in controlling cost.
- Help in preparation of flexible budgets.

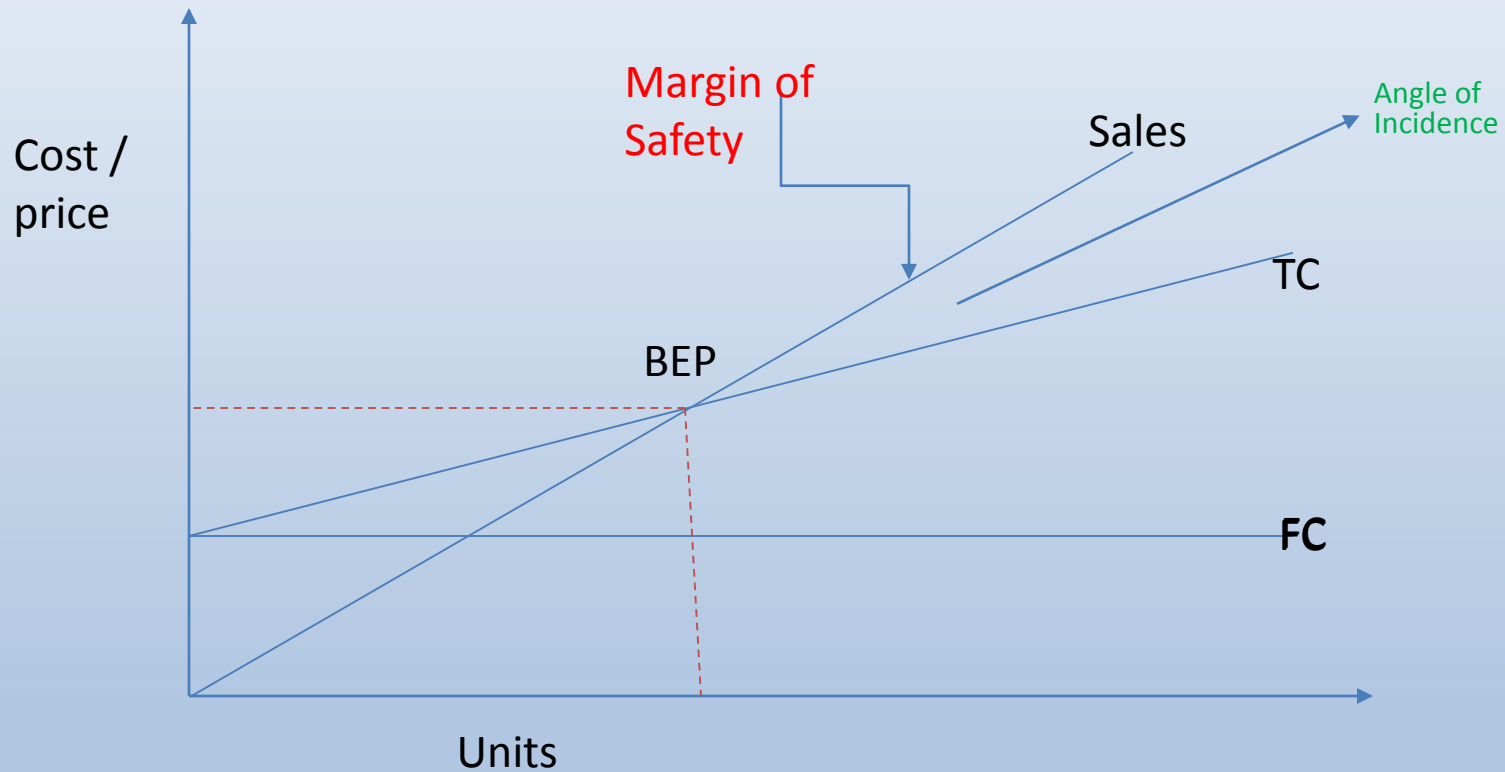
Limitations of BEP analysis

- Problems in identifying fixed and variable costs.
- Fixed cost not always fixed.
- Unit selling price not always constant.
- Not suitable for a multi-product firm.
- Ignoring the influence of other factors on cost and profit.

BEP Chart



BEP Chart



Service Entity

1. $BEP = \frac{FC}{CM\ Ratio}$ (amount)
2. $CM\ Ratio = \frac{CM}{Sales}$ or *Service Revenue* $= \frac{Sales - VC}{Sales}$ (%)
3. $Desired\ Sales\ or\ Service\ Revenue = \frac{FC + Targeted\ Profit}{CM\ Ratio}$
4. Margin of Safety : Total Sales – BE sales

October-2017

5/9

2(b)

(i) Safa Bank Ltd.

$$\begin{aligned}\text{Contribution Margin Ratio} &= \frac{\text{Contribution Margin}}{\text{Sales}} \\ &= \frac{\text{Sales} - \text{Variable Cost}}{\text{Sales}} \\ &= \frac{600000 - 480000}{600000} \\ &= \frac{120000}{600000} \\ &= 0.20 \\ &= 20\%\end{aligned}$$

$$\begin{aligned}\therefore \text{Break-even Point} &= \frac{\text{Fixed Cost}}{\text{Contribution Margin Ratio}} \\ &= \frac{600000}{0.20} \\ &= \text{Tk. } 300000. \text{ Ans.}\end{aligned}$$

$$\begin{aligned}\text{And Margin of Safety} &= \text{Tk. } (600000 - 300000) \\ &= \text{Tk. } 300000. \text{ Ans.}\end{aligned}$$

Thus for,

Morwa Bank Ltd.

$$\begin{aligned}\text{Contribution Margin Ratio} &= \frac{600000 - 400000}{600000} \\ &= \frac{200000}{600000} \\ &= 0.33 \text{ (approx.)} \\ &= 33\% \text{ (approx.)}\end{aligned}$$

$$\therefore \text{Break-even Point} = \frac{140000}{0.33} = \text{Tk. } 424,242 \text{ (approx.)}$$

$$\begin{aligned}\text{And Margin of Safety} &= \text{Tk. } (600000 - 424242) \\ &= \text{Tk. } 175,758\end{aligned}$$

6/9

(ii) Desired Service Revenue:

For Safa Bank Ltd.,

$$= \frac{\text{Fixed Cost} + \text{Targeted Profit}}{\text{Contribution Margin Ratio}}$$

$$= \frac{60,000 + 25,000}{0.20}$$

$$= \frac{85,000}{0.20}$$

$$= \text{Tk. } 425,000. \quad \underline{\text{Ans.}}$$

For Marwa Bank Ltd.,

$$= \frac{140,000 + 25,000}{0.33}$$

$$= \frac{165,000}{0.33}$$

$$= \text{Tk. } 500,000 \quad \text{Ans.}$$

(iii) Since Margin of safety of Safa Bank is higher than that of Marwa Bank, Safa Bank will earn greater profits in the given conditions.

779
April - 2016

2(b)

(i) We know,

$$\text{Break-even Point} = \frac{\text{Fixed Cost}}{\text{Contribution Margin Ratio}}$$

$$\text{And, Contribution Margin Ratio} = \frac{\text{Sales} - \text{Variable Cost}}{\text{Sales}}$$

AB Bank Ltd

$$\begin{aligned} \text{Contribution Margin Ratio: } & \frac{150000 - 120000}{150000} \\ &= \frac{30000}{150000} \\ &= 0.20 \\ &= 20\% \end{aligned}$$

$$\begin{aligned} \therefore \text{Break-even Point: } & \frac{15000}{0.20} \\ &= \underline{\underline{\text{Tk. } 75000}} \end{aligned}$$

margin of safety :

$$\begin{aligned} & \text{Service Revenue} \\ & \quad - \text{Break-even point} \\ &= 150000 - 75000 \\ &= \underline{\underline{\text{Tk. } 75000}} \end{aligned}$$

CD Bank Ltd.

$$\begin{aligned} & \frac{150000 - 100000}{150000} \\ &= \frac{50000}{150000} \\ &= 0.33 \text{ (approx)} \\ &= 33\% \end{aligned}$$

$$\begin{aligned} & \frac{35000}{0.33} \\ &= \underline{\underline{\text{Tk. } 106060 \text{ (approx)}}} \end{aligned}$$

Ans.

Service Revenue
- Break-even point

$$\begin{aligned} & 150000 - 106060 \\ &= \underline{\underline{\text{Tk. } 43940}} \end{aligned}$$

Ans.

(ii) we know,

$$\text{desired Sales} = \frac{\text{Fixed Cost} + \text{Targeted Profit}}{\text{Contribution Margin Ratio}}$$

AB Bank Ltd.

$$\begin{aligned} \text{desired service Revenue} &= \frac{15000 + 25000}{0.20} \\ &= \frac{40000}{0.20} \\ &= \text{Tk. } 200000 \end{aligned}$$

CD Bank Ltd.

$$\begin{aligned} &= \frac{35000 + 25000}{0.33} \\ &= \frac{60000}{0.33} \\ &= \text{Tk. } 181,818 \end{aligned}$$

(iii) Since margin of safety of AB Bank Ltd. is higher than that of CD Bank Ltd., AB Bank Ltd. will earn greater profits in the given conditions.

Manufacturing concerns

1. $\text{BEP (in units)} = \frac{FC}{CM \text{ per unit}}$
2. $\text{CM per unit} = \text{Selling price per unit} - \text{VC per unit}$
3. $\text{CM Ratio or PV Ratio} = \frac{CM \text{ per unit}}{\text{Selling Price per unit}} (\%)$
4. $\text{BEP (in Taka)} = \frac{FC}{CM \text{ Ratio}}$
5. $\text{Desired Sales} = \frac{FC + \text{Targeted Profit}}{CM \text{ Ratio}}$
6. $\text{Margin of Safety : Total Sales} - \text{BE sales}$

April 2019: 2(b)
i_)

	Factory A	Factory B
BEP in units:	$\frac{800000}{200-150} = 16000$	$\frac{1000000}{200-140} = 16667$
PV Ratio :	$\frac{200-150}{200} = 0.25 = 25\%$	$\frac{200-140}{200} = 0.3 = 30\%$
BEP in Taka :	$\frac{800000}{25\%} = \text{Tk. } 3200000$	$\frac{1000000}{30\%} = \text{Tk. } 33,33,333 \text{ (approx.)}$
Cash BEP in units :	$\frac{800000 - 120000}{200 - 150} = 13600$	$\frac{1000000 - 110000}{200 - 150} = 14833$
Cash BEP in Taka :	$\frac{800000 - 120000}{25\%} = \text{Tk. } 27,20,000$	
Margin of Safety :	$60000 - 16000 = 44000 \text{ units}$	
	Total Sales = $60000 \times 200 = \text{Tk. } 12000000$ BEP Sales = $\text{Tk. } 3200000$ MS = $12000000 - 3200000 = \text{Tk. } 88,00,000$	

April 2019: 2(b)
ii_)

	Factory A	Factory B
Now, VC Per unit :	$150 + 150 \times 10\% = \text{Tk. } 165$	$140 + 140 \times 15\% = \text{Tk } 161$
BEP in units:	$\frac{800000}{200 - 165} = 22857$	
PV Ratio :	$\frac{200 - 165}{200} = 0.175 = 17.50\%$	
BEP in Taka :	$\frac{800000}{17.50\%} = \text{Tk. } 4571428.57$	
Previous Profit	Total Sales – Total VC – FC = $60000 \times 200 - 60000 \times 150 - 800000 = \text{Tk. } 22,00,000$	
Present Profit	Total Sales – Total VC – FC = $60000 \times 200 - 60000 \times 165 - 800000 = \text{Tk. } 13,00,000$	
Consequences :	Profit decreases; BEP (units) increases; BEP (Taka) increases	

April 2019: 2(b)
iii_)

	Factory A	Factory B
Now, selling Price Per unit :	$200 - 200 \times 10\% = \text{Tk. } 180$	$200 - 200 \times 10\% = \text{Tk. } 180$
BEP in units:	$\frac{800000}{180 - 150} = 26667$	
PV Ratio :	$\frac{180 - 150}{180} = 0.1667 = 16.67\%$	
BEP in Taka :	$\frac{800000}{16.67\%} = \text{Tk. } 47,99,040 \text{ (approx.)}$	
iv)		
Profit :	$\text{Total Sales} - \text{Total VC} - \text{FC} = 50000 \times 200 - 50000 \times 150 - 800000 = \text{Tk. } 17,00,000$	

October 2019: 1(c)

	Here, Selling Price per unit is Tk. 500; VC per unit is Tk. 200; and FC is Tk. 1,10,000	
i)	BEP in units:	$\frac{110000}{500-200} = 367$
	PV Ratio :	$\frac{500-200}{500} = 0.60 = 60\%$
	BEP in Taka :	$\frac{110000}{60\%} = \text{Tk. } 183333 \text{ (approx.)}$
ii)	Profit :	Total Sales – Total VC – FC = $500 \times 500 - 500 \times 200 - 110000 = \text{Tk. } 40,000$
iii)	Desired Sales :	$\frac{FC + Targeted Profit}{CM Ratio} = \frac{110000 + 5000}{60\%} = \text{Tk. } 191666.67 \text{ (approx.)}$
iv)	Margin of Safety:	Now, Total Sales = $400 \times 500 = \text{Tk. } 200000$; BEP sales = Tk. 183333 MS = Tk. 16667

April 2018: 2(b)

Here, Selling Price per unit = $2350000/5000 = \text{Tk.}470$; VC per unit = $1300000/5000 = \text{Tk. } 260$; and FC is Tk. 7,35,000		
i)	PV Ratio :	$\frac{470-260}{470} = 0.45 = 45\% \text{ (approx.)}$
	BEP in Taka :	$\frac{735000}{45\%} = \text{Tk. } 16,33,33 \text{ (approx.)}$
	Margin of Safety:	$2350000 - 1633333 = \text{Tk. } 7,16,667$
ii)	Now VC per unit : $260 + 35 = \text{Tk. } 295$	
	PV Ratio :	$\frac{470-295}{470} = 0.37 = 37\% \text{ (approx.)}$
	BEP in units:	$\frac{735000}{470-295} = 4200$
iii)	Now, VC per unit = $260 - 260 \times 50\% = \text{Tk. } 130$; FC = $735000 + 735000 \times 80\% = \text{Tk. } 13,23,000$	
	PV Ratio :	$\frac{470-130}{470} = 0.72 = 72\% \text{ (approx.)}$
	Desired Sales :	$\frac{FC + Targeted Profit}{CM Ratio} = \frac{1323000 + 315120}{72\%} = \text{Tk.}22,75,167 \text{ (approx.)}$
	Desired Sales in units :	$2275167/470 = 4841$

Here, Selling Price per unit = Tk. 500; Material cost per unit : Tk. 100 Labor cost per unit : Tk. 160 Variable OH per unit : Tk. 120 (75% of labor cost) As such, VC per unit = 100 + 160 + 120 = Tk. 380; and FC is Tk. 4,80,000		
i)	BEP in units:	$\frac{480000}{500-380} = 4000$
	PV Ratio :	$\frac{500-380}{500} = 0.24 = 24\% \text{ (approx.)}$
	BEP in Taka :	$\frac{480000}{24\%} = \text{Tk. } 20,00,000$
ii)	Desired Sales :	$\frac{FC + Targeted Profit}{CM Ratio} = \frac{480000 + 250000}{24\%} = \text{Tk. } 30,41,667 \text{ (approx.)}$
	Desired Sales in units :	$3041667 / 500 = 6083$

Break Even

Oct' 2018

2(b)

- (i)
- | | |
|---|---------|
| Material per unit | Tk. 100 |
| Labor per unit | Tk. 160 |
| Variable Overhead per unit
(75% of Labor cost) | Tk. 120 |

Variable cost per unit: Tk. 380

Say, 'x' units to be sold at Break Even.

Then at B.E., Total Sales = Tk. 500x

Total VC = Tk. 380x

Total FC = Tk. 480,000

$$\therefore 500x - 380x - 480000 = 0$$

$$\text{or, } 120x = 480000$$

$$\therefore x = 4000$$

$\therefore$ BEP sales in units = 4000 units.

* BEP sales in amount = Tk. 500 $\times$ 4000 = Tk. 2,000,000

(ii) Suppose 'z' unit to be sold to make a profit of Tk. 250,000

$$\therefore 500z - 380z - 480000 = 250000$$

$$\text{or, } 120z = 250000 + 480000$$

$$\text{or, } 120z = 730000$$

$$\therefore z = 6083 \text{ (approx.)}$$

$\therefore$ In units = 6083 units (approx.)

* In amount = 500 $\times$ 6083 = Tk. 3,041,500

(iii) Suppose, 'p' units to be sold to make a profit of Tk. 150,000

$$\therefore 500P - 380P - 480,000 = 150,000$$

$$\text{or, } 120P = 150,000 + 480,000$$

$$\text{or, } 120P = 630,000$$

$$\therefore P = 5250$$

$$\therefore \text{In units} = 5250 \text{ units}$$

$$\& \text{In amount} = 500 \times 5250 = \text{Tk. } 2,625,000$$

$$\therefore \text{Margin of safety in units} = 5250 - 4000$$

$$= 1250 \text{ units}$$

from (i)
BEP sales
4000 units

$$\text{Margin of safety in amount} = 2,625,000 - 2,000,000$$

$$= \text{Tk. } 6,25,000$$

from (i) BEP
sales Tk. 2,000,000

(iv) Here, Total Sales = Tk. 2,500,000

$$\therefore \text{Units sold} = 2,500,000 / 500 = 5000 \text{ units}$$

$$\therefore \text{Total VC} = 380 \times 5000 = \text{Tk. } 1,900,000$$

$$\begin{aligned} \therefore \text{Total Cost} &= \text{Total VC} + \text{FC} \\ &= 1,900,000 + 480,000 \\ &= \text{Tk. } 2,380,000 \end{aligned}$$

$$\begin{aligned} \therefore \text{Profit} &= 2,500,000 - 2,380,000 \\ &= \text{Tk. } 120,000 \text{ Ans.} \end{aligned}$$

(v) New selling price per unit = $500 - 50 = \text{Tk. } 450$

Suppose, 'y' units to be sold at B.E.

$$\therefore 450y - 380y - 480,000 = 0$$

$$\text{or, } 70y = 480,000$$

$$\therefore y = 6857 \text{ (approx.)}$$

$$\therefore \text{BEP sales in units} = 6857 \text{ units (approx.)}$$

$$\& \text{BEP sales in amount} = 450 \times 6857$$

$$= \text{Tk. } 3,085,650$$

Break Even

April-2018

2(h)

(i)

$$\begin{aligned}\text{Selling price per unit} &= \frac{\text{Sales}}{\text{units sold}} \\ &= \frac{2350000}{5000} \\ &= \text{Tk. } 470\end{aligned}$$

$$\begin{aligned}\text{Variable Cost per unit} &= \frac{\text{Variable Expenses}}{\text{Units sold}} \\ &= \frac{1300000}{5000} \\ &= \text{Tk. } 260\end{aligned}$$

Suppose, 'x' units to be sold at Break-even.

Then, at Break-even,

$$\text{Total Sales} = x \times 470 = \text{Tk. } 470x$$

$$\text{Variable Cost} = x \times 260 = \text{Tk. } 260x$$

$$\text{Fixed Cost} = \text{Tk. } 735000$$

$$\therefore 470x - 260x - 735000 = 0$$

$$\text{or, } 210x = 735000$$

$$\therefore x = 3500$$

$$\therefore \text{Break-even-Point in Taka will be } (3500 \times 470) = \text{Tk. } 1645,000$$

$$\begin{aligned}\text{and Margin of Safety} &= 2350000 - 1645000 \\ &= \text{Tk. } 705,000.\end{aligned}$$

$$\begin{aligned}\text{Ans. : Break-even-Point} &= \text{Tk. } 1645,000; \\ \text{Margin of Safety} &= \text{Tk. } 705,000.\end{aligned}$$

$$(ii) \text{ Now, Variable Cost per unit} = 260 + 35 = \text{Tk. } 295$$

$$\therefore \text{Total Variable Cost} = 5000 \times 295 = \text{Tk. } 14,75,000$$

$$\text{Contribution Margin Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}}$$

2/9

$$\begin{aligned}
 &= \frac{\text{Sales} - \text{Variable Cost}}{\text{Sales}} \\
 &= \frac{2350000 - 1475000}{2350000} \\
 &= \frac{875000}{2350000} \\
 &= 0.37 \text{ (approx.)}
 \end{aligned}$$

Suppose, 'Q' units to be sold at Break-even-Point.

∴ At Break-even,

$$\text{Total Sales} = \text{Tk. } 470Q$$

$$\text{Variable Cost} = \text{Tk. } 295Q$$

$$\text{and Fixed Cost} = \text{Tk. } 7,35,000$$

$$\therefore 470Q - 295Q - 735000 = 0$$

$$\text{or, } 175Q = 735000$$

$$\therefore Q = 4200$$

Ans. ∴ Contribution Margin Ratio 0.37 or 37% (approx.)
and Break-even-Point in number 4200 units.

$$\begin{aligned}
 \text{(iii) Here, Variable Cost} &= 1300000 - (1300000 \times 50\%) \\
 &= 1300000 - 650000 \\
 &= \text{Tk. } 650,000
 \end{aligned}$$

$$\begin{aligned}
 \text{Fixed Cost} &= 7,35,000 + (735000 \times 80\%) \\
 &= 735000 + 5,88,000 \\
 &= \text{Tk. } 13,23,000
 \end{aligned}$$

$$\therefore \text{New Variable Cost per unit} = \frac{650000}{5000} = \text{Tk. } 130$$

and selling price per unit will remain same i.e. Tk. 470.

Suppose, 'Y' units to be sold to earn net income of Tk. 315120.

$$\therefore 470Y - 130Y - 1323000 = 315120$$

$$\text{or, } 340Y = 315120 + 1323000$$

$$\text{or, } 340Y = 16,38,120$$

$$\therefore Y = 4818 \quad \text{Ans.}$$

2 (c).

April - 2017 ^{3/9}

(i)

Figure i

Units Sold (1)	Selling price per unit (2) (TK)	Variable Cost per unit (3) (TK)	Fixed Cost (4) (TK)	Total Sales (5) = 1x2 (TK)	Total Variable Cost (6) = 1x3 (TK)	Profit/Loss (7) = 5-6-4 (TK)
500	100	60	4000	50000	30000	16000
200	100	60	4000	20000	12000	4000
100	100	60	4000	10000	6000	0

∴ projected profit will be TK. 16000, TK. 4000 and TK. nil respectively. Ans.

(ii) Suppose, 'a' units to be sold to earn profit of TK. 5000.

Then, Total sales = TK. 100a

Total Variable Cost = TK. 60a

$$\therefore 100a - 60a - 4000 = 5000$$

$$\text{or, } 40a = 5000 + 4000$$

$$\text{or, } 40a = 9000$$

$$\therefore a = 225$$

∴ Sales Revenue will be TK. $(225 \times 100) = \text{TK. } 22500$. Ans.

(iii) Here, Sales TK. 30000

$$\therefore \text{Units sold} = \frac{30000}{100} = 300$$

$$\therefore \text{Total Variable Cost} = 300 \times 60 = \text{TK. } 18000$$

$$\begin{aligned} \therefore \text{Projected Profit} &= \text{Sales} - \text{Variable Cost} - \text{Fixed Cost} \\ &= 30000 - 18000 - 4000 \\ &= \text{TK. } 8000 \quad \text{Ans.} \end{aligned}$$

1/9

(iv) From (i) we get, Break-even-Point unit 100

$\therefore$ Margin of safety in units will be $(400 - 100)$
or 300. Ans.

And Margin of safety in value $= 300 \times 100$
 $= \text{Tk. } 30000$ Ans.

(v) Now, selling price per unit $= 100 - (100 \times 10\%)$
 $= 100 - 10$
 $= \text{Tk. } 90$

Suppose, 'Q' units to be sold for Break-even-Point.

$$\therefore 90Q - 60Q - 4000 = 0$$

$$\text{or, } 30Q = 4000$$

$$\therefore Q = \frac{4000}{30}$$

$$= 133 \text{ (approx.)}$$

Here, Sales $= 90 \times Q$
 $= \text{Tk. } 90Q$
Variable Cost $= 60 \times Q$
 $= \text{Tk. } 60Q$
Fixed Cost $= \text{Tk. } 4000$

Break-even-Point in Taka will be (133×90)
or $\text{Tk. } 11,970$ (approx.)
Ans.

October-2017

5/9

2(b)

(i) Safa Bank Ltd.

$$\begin{aligned}\text{Contribution Margin Ratio} &= \frac{\text{Contribution Margin}}{\text{Sales}} \\ &= \frac{\text{Sales} - \text{Variable Cost}}{\text{Sales}} \\ &= \frac{600000 - 480000}{600000} \\ &= \frac{120000}{600000} \\ &= 0.20 \\ &= 20\%\end{aligned}$$

$$\begin{aligned}\therefore \text{Break-even Point} &= \frac{\text{Fixed Cost}}{\text{Contribution Margin Ratio}} \\ &= \frac{600000}{0.20} \\ &= \text{Tk. } 300000. \text{ Ans.}\end{aligned}$$

$$\begin{aligned}\text{And Margin of Safety} &= \text{Tk. } (600000 - 300000) \\ &= \text{Tk. } 300000. \text{ Ans.}\end{aligned}$$

Thus for,

Morwa Bank Ltd.

$$\begin{aligned}\text{Contribution Margin Ratio} &= \frac{600000 - 400000}{600000} \\ &= \frac{200000}{600000} \\ &= 0.33 \text{ (approx.)} \\ &= 33\%\end{aligned}$$

$$\therefore \text{Break-even Point} = \frac{140000}{0.33} = \text{Tk. } 424,242 \text{ (approx.)}$$

$$\begin{aligned}\text{And Margin of Safety} &= \text{Tk. } (600000 - 424242) \\ &= \text{Tk. } 175,758\end{aligned}$$

6/9

(ii) Desired Service Revenue:

For Safa Bank Ltd.,

$$= \frac{\text{Fixed Cost} + \text{Targeted Profit}}{\text{Contribution Margin Ratio}}$$

$$= \frac{60,000 + 25,000}{0.20}$$

$$= \frac{85,000}{0.20}$$

$$= \text{Tk. } 425,000. \quad \underline{\text{Ans.}}$$

For Marwa Bank Ltd.,

$$= \frac{140,000 + 25,000}{0.33}$$

$$= \frac{165,000}{0.33}$$

$$= \text{Tk. } 500,000 \quad \text{Ans.}$$

(iii) Since Margin of safety of Safa Bank is higher than that of Marwa Bank, Safa Bank will earn greater profits in the given conditions.

779
April - 2016

2(b)

(i) We know,

$$\text{Break-even Point} = \frac{\text{Fixed Cost}}{\text{Contribution Margin Ratio}}$$

$$\text{And, Contribution Margin Ratio} = \frac{\text{Sales} - \text{Variable Cost}}{\text{Sales}}$$

AB Bank Ltd

$$\begin{aligned} \text{Contribution Margin Ratio: } & \frac{150000 - 120000}{150000} \\ &= \frac{30000}{150000} \\ &= 0.20 \\ &= 20\% \end{aligned}$$

$$\begin{aligned} \therefore \text{Break-even Point: } & \frac{15000}{0.20} \\ &= \underline{\underline{\text{Tk. } 75000}} \end{aligned}$$

margin of safety :

$$\begin{aligned} & \text{Service Revenue} \\ & \quad - \text{Break-even point} \\ &= 150000 - 75000 \\ &= \underline{\underline{\text{Tk. } 75000}} \end{aligned}$$

CD Bank Ltd.

$$\begin{aligned} & \frac{150000 - 100000}{150000} \\ &= \frac{50000}{150000} \\ &= 0.33 \text{ (approx)} \\ &= 33\% \end{aligned}$$

$$\begin{aligned} & \frac{35000}{0.33} \\ &= \underline{\underline{\text{Tk. } 106060 \text{ (approx)}}} \end{aligned}$$

Ans.

Service Revenue
- Break-even point

$$\begin{aligned} & 150000 - 106060 \\ &= \underline{\underline{\text{Tk. } 43940}} \end{aligned}$$

Ans.

(ii) we know,

$$\text{desired Sales} = \frac{\text{Fixed Cost} + \text{Targeted Profit}}{\text{Contribution Margin Ratio}}$$

AB Bank Ltd.

$$\begin{aligned} \text{desired service} & \\ \text{Revenue : } & \frac{15000 + 25000}{0.20} \\ & = \frac{40000}{0.20} \\ & = \text{Tk. } 200000 \end{aligned}$$

CD Bank Ltd.

$$\begin{aligned} & \frac{35000 + 25000}{0.33} \\ & = \frac{60000}{0.33} \\ & = \text{Tk. } 181,818 \end{aligned}$$

(iii) Since margin of safety of AB Bank Ltd. is higher than that of CD Bank Ltd., AB Bank Ltd. will earn greater profits in the given conditions.

2(17)

March - 2015

9/9

(i) Selling Price per unit : Tk. 3

Variable Cost per unit : $7500 \div 5000$

= Tk. 1.5

Suppose, 'Q' units to be sold for Break-even Point.

$$\therefore 3Q - 1.5Q - 4500 = 0$$

$$\text{or, } 1.5Q = 4500$$

$$\therefore Q = \frac{4500}{1.5}$$
$$= 3000$$

$$\left\{ \begin{array}{l} \text{Here,} \\ \text{Total Sales} = 3 \times Q \\ \quad = \text{Tk. } 3Q \\ \text{Total Variable} \\ \text{Cost} = 1.5 \times Q \\ \quad = \text{Tk. } 1.5Q \\ \text{Fixed Cost} = \text{Tk. } 4500 \end{array} \right.$$

$\therefore$ Break-even point in units 3000 Ans

(ii) margin of safety in units : $5000 - 3000$
= 2000 units.

margin of safety in Taka = 2000×3
= Tk. 6000. Ans.

(iii) ~~desired sales~~ = ~~Fixed Cost + Targeted Profit~~

Suppose, 'x' units to be sold to earn profit of Tk. 6000.

$$\therefore 3x - 1.5x - 4500 = 6000$$

$$\text{or, } 1.5x = 6000 + 4500$$

$$\therefore x = \frac{10500}{1.5}$$
$$= 7000$$

$$\left\{ \begin{array}{l} \text{Here,} \\ \text{Total Sales} = \text{Tk. } 3x \\ \text{Variable Cost} = \text{Tk. } 1.5x \\ \text{Fixed Cost} = \text{Tk. } 4500 \\ \text{Targeted Profit} = \text{Tk. } 6000 \end{array} \right.$$

$\therefore$ Required sales : 7000 units at ~~Tk.~~

Tk. (7000×3)

or Tk. 21000.

Ans..